

Preparación:

```
[vrodriquez@contenedores ~]$ docker info
Client: Docker Engine - Community
Version: 29.4.1
Context: default
Debug Mode: false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.33.0
    Path: /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version: v5.1.3
    Path: /usr/libexec/docker/cli-plugins/docker-compose

Server:
Containers: 2
  Running: 0
  Paused: 0
  Stopped: 2
Images: 1
Server Version: 29.4.1
Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
  /etc/cdi
```

```
/etc/cdi
/var/run/cdi
Swarm: inactive
Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version: 77c84241c7cbdd9b4eca2591793e3d4f4317c590
runc version: v1.3.5-0-g488fc13e
init version: de40ad0
Security Options:
  seccomp
    Profile: builtin
  cgroupns
Kernel Version: 5.14.0-611.5.1.el9_7.x86_64
Operating System: Rocky Linux 9.7 (Blue Onyx)
OSType: linux
Architecture: x86_64
CPUs: 1
Total Memory: 1.667GiB
Name: contenedores
ID: ade8ac7b-1b59-494b-a3ae-c81fb4fadd23
Docker Root Dir: /var/lib/docker
Debug Mode: false
Experimental: false
Insecure Registries:
  ::1/128
  127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables+firewalld
ReloadedAt: 2026-04-24T22:54:07-05:00
```

```
[vrodriquez@contenedores ~]$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
hello-world:latest	f9078146db2e	21.8kB	9.49kB	 Info →  In Use
lscr.io/linuxserver/webtop:latest	c18034f25b0a	3.32GB	924MB	
nginx:latest	6e23479198b9	237MB	65.8MB	

```
[vrodriquez@contenedores ~]$
```

Imagen:

```
[vrodriiguez@contenedores docker-lab]$ vim Dockerfile
[vrodriiguez@contenedores docker-lab]$ cat Dockerfile
FROM alpine:3.20
WORKDIR /app
RUN echo "Hola desde Docker" > /app/mensaje.txt
CMD ["cat", "/app/mensaje.txt"]

[vrodriiguez@contenedores docker-lab]$ docker build -t miapp:v1 .
[+] Building 8.8s (7/7) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 207B                             0.0s
=> [internal] load metadata for docker.io/library/alpine:3.20   3.7s
=> [internal] load .dockerignore                                0.1s
=> => transferring context: 2B                                    0.0s
=> [1/3] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc 2.4s
=> => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc 0.1s
=> => sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd9f98b50d2e1b96130471 3.63MB / 3.63MB 2.0s
=> => extracting sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd9f98b50d2e1b96130471 0.2s
=> [2/3] WORKDIR /app                                           0.3s
=> [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt       0.7s
=> exporting to image                                           1.0s
=> => exporting layers                                           0.6s
=> => exporting manifest sha256:1980c3128d77e9219e26091dfcae16f472425b538083b473bb64a0bf9697e73b 0.0s
=> => exporting config sha256:4762603418f892aba27e777158f00c2584ea7132e3949ef4539c57d9acd7f7c 0.0s
=> => exporting attestation manifest sha256:3a2b0c75fabfd900e7d36c ff33a067c04ca9391101d0da5eee127c589bf0c088 0.1s
=> => exporting manifest list sha256:421a47947aa00056b6275ccd9c335fafad2b50d965d148ebac7cf4fff0831c49 0.0s
=> => naming to docker.io/library/miapp:v1                      0.0s
=> => unpacking to docker.io/library/miapp:v1                    0.1s
[vrodriiguez@contenedores docker-lab]$ docker images

INFO → U In Use
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
hello-world:latest                  f9078146db2e                      21.8kB      9.49kB        U
lscr.io/linuxserver/webtop:latest   c18034ff25b0a                    3.32GB      924MB
miapp:v1                            421a47947aa0                      11.8MB      3.63MB
nginx:latest                        6e23479198b9                      237MB       65.8MB
[vrodriiguez@contenedores docker-lab]$
```

```
[vrodriiguez@contenedores docker-lab]$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1                            421a47947aa0                      11.8MB      3.63MB
[vrodriiguez@contenedores docker-lab]$ docker run --rm miapp:v1
Hola desde Docker
[vrodriiguez@contenedores docker-lab]$
```

```
[vrodriiguez@contenedores docker-lab]$ docker tag miapp:v1 miapp:latest
[vrodriiguez@contenedores docker-lab]$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest                        421a47947aa0                      11.8MB      3.63MB
miapp:v1                            421a47947aa0                      11.8MB      3.63MB
[vrodriiguez@contenedores docker-lab]$
```

```
[vrodriiguez@contenedores docker-lab]$ docker image inspect miapp:v1
[
  {
    "Id": "sha256:421a47947aa00056b6275ccd9c335fafad2b50d965d148ebac7cf4fff0831c49",
    "RepoTags": [
      "miapp:latest",
      "miapp:v1"
    ],
    "RepoDigests": [
      "miapp@sha256:421a47947aa00056b6275ccd9c335fafad2b50d965d148ebac7cf4fff0831c49"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-24T23:36:02.464121698-05:00",
    "Config": {
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
      ],
      "Cmd": [
        "cat",
        "/app/mensaje.txt"
      ],
      "WorkingDir": "/app",
      "ArgsEscaped": true
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 3633527,
    "RootFS": {
      "Type": "layers",
      "Layers": [
        "sha256:08bc4e534116aa76b16015484b82eac51f9a593416feae9296c8a2d4bb7aa4a2",
        "sha256:6c968f9bba5ba883983078f13f4afff31c433cbf89e7b1b49ceadda389147570",
        "sha256:08d0b9e21656b5851d0d8914a396d5ce4cfe98f4d78891e2f1598b6722df01b0"
      ]
    }
  },

```

```

  },
  "Architecture": "amd64",
  "Os": "linux",
  "Size": 3633527,
  "RootFS": {
    "Type": "layers",
    "Layers": [
      "sha256:08bc4e534116aa76b16015484b82eac51f9a593416feae9296c8a2d4bb7aa4a2",
      "sha256:6c968f9bba5ba883983078f13f4afff31c433cbf89e7b1b49ceadda389147570",
      "sha256:08d0b9e21656b5851d0d8914a396d5ce4cfe98f4d78891e2f1598b6722df01b0"
    ]
  },
  "Metadata": {
    "LastTagTime": "2026-04-25T04:39:43.793601522Z"
  },
  "Descriptor": {
    "mediaType": "application/vnd.oci.image.index.v1+json",
    "digest": "sha256:421a47947aa00056b6275ccd9c335fafad2b50d965d148ebac7cf4fff0831c49",
    "size": 855
  },
  "Identity": {
    "Build": [
      {
        "Ref": "9f1lwq9akr58jil6s9jts22lw",
        "CreatedAt": "2026-04-24T23:36:03.557606249-05:00"
      }
    ]
  }
}
]
[vrodriiguez@contenedores docker-lab]$
```

```
[vrodriiguez@contenedores docker-lab]$ docker save -o miapp.tar miapp:v1
[vrodriiguez@contenedores docker-lab]$ ls -lh miapp.tar
-rw-r--r-- 1 vrodriiguez docker 3.5M Apr 24 23:41 miapp.tar
[vrodriiguez@contenedores docker-lab]$
```

```
[vrodriguez@contenedores docker-lab]$ docker rmi miapp:v1
Untagged: miapp:v1
[vrodriguez@contenedores docker-lab]$ docker load -i miapp.tar
Loaded image: miapp:v1
[vrodriguez@contenedores docker-lab]$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:latest          421a47947aa0      11.8MB      3.63MB
miapp:v1              421a47947aa0      11.8MB      3.63MB
[vrodriguez@contenedores docker-lab]$
```

```
[vrodriguez@contenedores docker-lab]$ docker rmi miapp:latest
Untagged: miapp:latest
[vrodriguez@contenedores docker-lab]$ docker images | grep miapp
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
miapp:v1              421a47947aa0      11.8MB      3.63MB
[vrodriguez@contenedores docker-lab]$
```

```
[vrodriguez@contenedores docker-lab]$ docker image prune -f
Total reclaimed space: 0B
[vrodriguez@contenedores docker-lab]$
```

Docker Container:

```
[vrodriquez@contenedores docker-lab]$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
124ca10f52f2	nginx:latest	"/docker-entrypoint..."	About a minute ago	Up About a minute	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
32e94bf0cc29	nginx:latest	"/docker-entrypoint..."	About a minute ago	Up About a minute	80/tcp

```
[vrodriquez@contenedores docker-lab]$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[vrodriquez@contenedores docker-lab]$
```

```
[vrodriquez@contenedores docker-lab]$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
124ca10f52f2	nginx:latest	"/docker-entrypoint..."	5 minutes ago	Up 5 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
32e94bf0cc29	nginx:latest	"/docker-entrypoint..."	5 minutes ago	Up 5 minutes	80/tcp
31add67fdf31	hello-world	"/hello"	46 minutes ago	Exited (0) 46 minutes ago	
7a5dfd47bc0d	hello-world	"/hello"	55 minutes ago	Exited (0) 55 minutes ago	

```
[vrodriquez@contenedores docker-lab]$
```

```
[vrodriquez@contenedores docker-lab]$ docker logs web2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/25 04:45:14 [notice] 1#1: using the "epoll" event method
2026/04/25 04:45:14 [notice] 1#1: nginx/1.29.8
2026/04/25 04:45:14 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/25 04:45:14 [notice] 1#1: OS: Linux 5.14.0-611.5.1.el9_7.x86_64
2026/04/25 04:45:14 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
2026/04/25 04:45:14 [notice] 1#1: start worker processes
2026/04/25 04:45:14 [notice] 1#1: start worker process 29
172.17.0.1 - - [25/Apr/2026:04:47:12 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/7.76.1" "-"
[vrodriquez@contenedores docker-lab]$
```

```
[vrodriquez@contenedores docker-lab]$ docker inspect web2
[
  {
    "Id": "124ca10f52f2bd4689dc1a99f2fe0d7d4bbb581d9631eb821d27661a9131668e",
    "Created": "2026-04-25T04:45:13.251063802Z",
    "Path": "/docker-entrypoint.sh",
    "Args": [
      "nginx",
      "-g",
      "daemon off;"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 38980,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-04-25T04:45:13.813247709Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:6e23479198b998e5e25921dff8455837c7636a6711a04a635cf1bb363d199dc",
    "ResolvConfPath": "/var/lib/docker/containers/124ca10f52f2bd4689dc1a99f2fe0d7d4bbb581d9631eb821d27661a9131668e/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/124ca10f52f2bd4689dc1a99f2fe0d7d4bbb581d9631eb821d27661a9131668e/hostname",
    "HostsPath": "/var/lib/docker/containers/124ca10f52f2bd4689dc1a99f2fe0d7d4bbb581d9631eb821d27661a9131668e/hosts",
    "LogPath": "/var/lib/docker/containers/124ca10f52f2bd4689dc1a99f2fe0d7d4bbb581d9631eb821d27661a9131668e/124ca10f52f2bd4689dc1a99f2fe0d7d4bbb581d9631eb821d27661a9131668e-json.log",
    "Name": "/web2",
    "RestartCount": 0,
    "Driver": "overlayfs",
    "Platform": "linux",
    "MountLabel": ""
  }
]
```

```
    "Gateway": "172.17.0.1",
    "IPAddress": "172.17.0.3",
    "MacAddress": "6e:44:ca:1f:a7:b8",
    "IPPrefixLen": 16,
    "IPv6Gateway": "",
    "GlobalIPv6Address": "",
    "GlobalIPv6PrefixLen": 0,
    "DNSNames": null
  }
},
"ImageManifestDescriptor": {
  "mediaType": "application/vnd.oci.image.manifest.v1+json",
  "digest": "sha256:296499281873ebe2636681c6957776d29a31858d93c3e39b654b5aade87ed70a",
  "size": 2290,
  "annotations": {
    "com.docker.official-images.bashbrew.arch": "amd64",
    "org.opencontainers.image.base.digest": "sha256:e18da95f66066b7c5fa31491b524e83121271eca59a3d140f4906c8d0a090367",
    "org.opencontainers.image.base.name": "debian:trixie-slim",
    "org.opencontainers.image.created": "2026-04-22T01:21:41Z",
    "org.opencontainers.image.revision": "71081b25390771f6b1275ddf7c73c965f304493f",
    "org.opencontainers.image.source": "https://github.com/nginx/docker-nginx.git#71081b25390771f6b1275ddf7c73c965f304493f:mainline/debian",
    "org.opencontainers.image.url": "https://hub.docker.com/_/nginx",
    "org.opencontainers.image.version": "1.29.8"
  },
  "platform": {
    "architecture": "amd64",
    "os": "linux"
  }
}
]
[vrodriquez@contenedores docker-lab]$ docker inspect web2 --format '{{.State.Status}}'
running
[vrodriquez@contenedores docker-lab]$
```

```
[vrodriquez@contenedores docker-lab]$ docker stop web2
web2
[vrodriquez@contenedores docker-lab]$ docker start web2
web2
[vrodriquez@contenedores docker-lab]$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
ES
124ca10f52f2   nginx:latest   "/docker-entrypoint.…"   7 minutes ago   Up 4 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   web2
32e94bf0cc29   nginx:latest   "/docker-entrypoint.…"   8 minutes ago   Up 8 minutes   80/tcp                    web1
1
[vrodriquez@contenedores docker-lab]$
```

```
[vrodriiguez@contenedores docker-lab]$ docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
Digest: sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
Status: Downloaded newer image for alpine:3.20
/ #
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.10
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit
[vrodriiguez@contenedores docker-lab]$ docker ps -a | grep alpine1
db5452493eca    alpine:3.20    "sh"                About a minute ago    Exited (0) 10 seconds ago
alpine1
```

```
[vrodriiguez@contenedores docker-lab]$ docker pause web1
web1
[vrodriiguez@contenedores docker-lab]$ docker unpause web1
web1
[vrodriiguez@contenedores docker-lab]$ docker inspect web1 --format '{{.State.Paused}}'
false
[vrodriiguez@contenedores docker-lab]$
```

```
[vrodriiguez@contenedores docker-lab]$ docker stop alpine1
alpine1
[vrodriiguez@contenedores docker-lab]$ docker rm alpine1
alpine1
[vrodriiguez@contenedores docker-lab]$ docker rm -f web1
web1
[vrodriiguez@contenedores docker-lab]$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
124ca10f52f2	nginx:latest	"/docker-entrypoint..."	10 minutes ago	Up 3 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
31add67fdf31	hello-world	"/hello"	52 minutes ago	Exited (0) 52 minutes ago	
7a5dfd47bc0d	hello-world	"/hello"	About an hour ago	Exited (0) About an hour ago	

```

reverent_golick
lucid_germain
[vrodriiguez@contenedores docker-lab]$
```

Volúmenes:

```
[vrodriiguez@contenedores docker-lab]$ docker volume create datos_nginx
datos_nginx
[vrodriiguez@contenedores docker-lab]$ docker volume ls
```

DRIVER	VOLUME NAME
local	datos_nginx

```
[vrodriiguez@contenedores docker-lab]$ docker volume inspect datos_nginx
[
  {
    "CreatedAt": "2026-04-24T23:56:41-05:00",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
    "Name": "datos_nginx",
    "Options": null,
    "Scope": "local"
  }
]
```

```
[vrodriiguez@contenedores docker-lab]$ docker run -d --name webvol -p 8081:80 -v datos_nginx:/usr/share/nginx/html nginx:latest
b381f9ac5e831c15af78ad8fcbdd66eba4f1e714c1a1512f86b2d9b84f262794f
[vrodriiguez@contenedores docker-lab]$ echo "<h1>Hola volumen</h1>" > index.html
[vrodriiguez@contenedores docker-lab]$ docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 22B (transferred 2.05kB) to webvol:/usr/share/nginx/html/index.html
[vrodriiguez@contenedores docker-lab]$ curl http://localhost:8081
<h1>Hola volumen</h1>
[vrodriiguez@contenedores docker-lab]$
```



```
[vrodriiguez@contenedores docker-lab]$ docker rm -f webvol
webvol
[vrodriiguez@contenedores docker-lab]$ docker volume rm datos_nginx
datos_nginx
[vrodriiguez@contenedores docker-lab]$ docker volume ls
DRIVER      VOLUME NAME
[vrodriiguez@contenedores docker-lab]$
```

Redes:

```
[vrodriiguez@contenedores docker-lab]$ docker network ls
NETWORK ID        NAME        DRIVER      SCOPE
fa55c6f9b159      bridge      bridge      local
73f4fc81641d      host        host        local
9c38296daf9d      none        null        local
[vrodriiguez@contenedores docker-lab]$ docker network create mi_red
16db3a600a1de9ad52a448e274294e7bf742a84ced2532d822e64c35fa397061
[vrodriiguez@contenedores docker-lab]$ docker network ls
NETWORK ID        NAME        DRIVER      SCOPE
fa55c6f9b159      bridge      bridge      local
73f4fc81641d      host        host        local
16db3a600a1d      mi_red      bridge      local
9c38296daf9d      none        null        local
[vrodriiguez@contenedores docker-lab]$ docker network inspect mi_red
[
  {
    "Name": "mi_red",
    "Id": "16db3a600a1de9ad52a448e274294e7bf742a84ced2532d822e64c35fa397061",
    "Created": "2026-04-25T00:06:51.233394231-05:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
  }
]
```



```

    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {},
    "Status": {
      "IPAM": {
        "Subnets": {
          "172.18.0.0/16": {
            "IPsInUse": 3,
            "DynamicIPsAvailable": 65533
          }
        }
      }
    }
  }
}
]
[vrodriguez@contenedores docker-lab]$ docker run -d --name caja1 --network mi_red alpine:3.20 sleep 300
6b09c2a2dfb3b4e63100e65bcc90a3a184ccc857553a247d1123d42ee0868d7c
[vrodriguez@contenedores docker-lab]$ docker run -d --name caja2 --network mi_red alpine:3.20 sleep 300
c31eb79a4645d24e2198a1f970872b7effa37128766b262e2bddfd76023e78a5
[vrodriguez@contenedores docker-lab]$ docker exec caja1 ping -c 2 caja2
PING caja2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.227 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.128 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.128/0.177/0.227 ms
[vrodriguez@contenedores docker-lab]$

```

```

[vrodriguez@contenedores docker-lab]$ docker network disconnect mi_red caja2
[vrodriguez@contenedores docker-lab]$ docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'
[vrodriguez@contenedores docker-lab]$

```

Docker Compose :

```

[vrodriguez@contenedores docker-lab]$ vim compose.yml
[vrodriguez@contenedores docker-lab]$ cat compose.yml
services:
  web:
    image: nginx:latest
    ports:
      - "8082:80"

[vrodriguez@contenedores docker-lab]$ docker compose up -d
[+] up 2/2
 ✓ Network docker-lab_default Created 0.3s
 ✓ Container docker-lab-web-1 Started 0.7s
[vrodriguez@contenedores docker-lab]$

```

```
[vrodriquez@contenedores docker-lab]$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED        STATUS        PORTS
docker-lab-web-1    nginx:latest      "/docker-entrypoint...." web         28 seconds ago Up 28 seconds 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
[vrodriquez@contenedores docker-lab]$ curl http://localhost:8082
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[vrodriquez@contenedores docker-lab]$
```

```
[vrodriquez@contenedores docker-lab]$ docker compose logs
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/04/25 05:14:51 [notice] 1#1: using the "epoll" event method
web-1 | 2026/04/25 05:14:51 [notice] 1#1: nginx/1.29.8
web-1 | 2026/04/25 05:14:51 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1 | 2026/04/25 05:14:51 [notice] 1#1: OS: Linux 5.14.0-611.5.1.el9_7.x86_64
web-1 | 2026/04/25 05:14:51 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:524288
web-1 | 2026/04/25 05:14:51 [notice] 1#1: start worker processes
web-1 | 2026/04/25 05:14:51 [notice] 1#1: start worker process 28
web-1 | 172.19.0.1 - - [25/Apr/2026:05:15:26 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/7.76.1" "-"
[vrodriquez@contenedores docker-lab]$
```

```
[vrodriquez@contenedores docker-lab]$ docker compose down
[+] down 2/2
✓ Container docker-lab-web-1 Removed 0.4s
✓ Network docker-lab_default Removed 0.3s
[vrodriquez@contenedores docker-lab]$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED        STATUS        PORTS
[vrodriquez@contenedores docker-lab]$
```

```
[vrodriquez@contenedores docker-lab]$ docker compose ps
NAME                IMAGE             COMMAND                  SERVICE    CREATED        STATUS        PORTS
docker-lab-web-1    nginx:latest      "/docker-entrypoint...." web         4 minutes ago Up 55 seconds
docker-lab-web-2    nginx:latest      "/docker-entrypoint...." web         4 minutes ago Up 4 minutes 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
docker-lab-web3-1   nginx:latest      "/docker-entrypoint...." web3        56 seconds ago Up 55 seconds 0.0.0.0:8083->80/tcp, [::]:8083->80/tcp
[vrodriquez@contenedores docker-lab]$
```

Dockerfile:

```
[vrodriiguez@contenedores docker-lab]$ mkdir dockerfile-test
[vrodriiguez@contenedores docker-lab]$ cd dockerfile-test
[vrodriiguez@contenedores dockerfile-test]$ echo "contenido local" > archivo.txt
[vrodriiguez@contenedores dockerfile-test]$ vim Dockerfile
[vrodriiguez@contenedores dockerfile-test]$ cat Dockerfile
FROM alpine:3.20
LABEL autor="prueba"
ARG VERSION=1.0
ENV APP_ENV=dev
WORKDIR /app
COPY archivo.txt /app/archivo.txt
RUN echo "version=$VERSION" > /app/version.txt
ADD archivo.txt /app/archivo2.txt
VOLUME ["/data"]
CMD ["sh", "-c", "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"]
[vrodriiguez@contenedores dockerfile-test]$
```

```
[vrodriiguez@contenedores dockerfile-test]$ docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 3.2s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 413B                                0.0s
=> [internal] load metadata for docker.io/library/alpine:3.20     0.1s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc 0.1s
=> => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc 0.1s
=> [internal] load build context                                   0.2s
=> => transferring context: 113B                                     0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> [3/5] COPY archivo.txt /app/archivo.txt                         0.1s
=> [4/5] RUN echo "version=2.0" > /app/version.txt                 0.7s
=> [5/5] ADD archivo.txt /app/archivo2.txt                         0.4s
=> exporting to image                                             1.3s
=> => exporting layers                                              0.7s
=> => exporting manifest sha256:0b88281dc53595e1ca7a5d6ba1c868f32b4fafc0b1ef32c137ab314d8765fcb3 0.0s
=> => exporting config sha256:cf9eb082d584d78d1f23f9d3c869154008f5f741a7324fdbaf9a6389ae74fd 0.0s
=> => exporting attestation manifest sha256:86b7153983e06a01ef27a46cb1cb7bbf755e2e7ea597c221b6edbb1103d9cc56 0.1s
=> => exporting manifest list sha256:0a3c72a6b5cbc615bf618747b05d2205c8bc4c5583eb4cef95453410a6f652fd 0.0s
=> => naming to docker.io/library/dockerfile-demo:latest          0.0s
=> => unpacking to docker.io/library/dockerfile-demo:latest       0.4s
[vrodriiguez@contenedores dockerfile-test]$ docker run --rm dockerfile-demo
APP_ENV=dev
contenido local
version=2.0
[vrodriiguez@contenedores dockerfile-test]$
```

```
[vrodriiguez@contenedores dockerfile-test]$ docker image inspect dockerfile-demo
[
  {
    "Id": "sha256:0a3c72a6b5cbc615bf618747b05d2205c8bc4c5583eb4cef95453410a6f652fd",
    "RepoTags": [
      "dockerfile-demo:latest"
    ],
    "RepoDigests": [
      "dockerfile-demo@sha256:0a3c72a6b5cbc615bf618747b05d2205c8bc4c5583eb4cef95453410a6f652fd"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-25T00:27:25.188432471-05:00",
    "Config": {
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "APP_ENV=dev"
      ],
      "Cmd": [
        "sh",
        "-c",
        "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"
      ],
      "Volumes": {
        "/data": {}
      },
      "WorkingDir": "/app",
      "Labels": {
        "autor": "prueba"
      },
      "ArgsEscaped": true
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 3635424,
    "RootFS": {
      "Type": "layers",

```

```
[vrodriiguez@contenedores dockerfile-test]$ docker image inspect dockerfile-demo --format '{{json .Config.Labels}}'
{"autor":"prueba"}
[vrodriiguez@contenedores dockerfile-test]$ docker image inspect dockerfile-demo --format '{{json .Config.Env}}'
["PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin","APP_ENV=dev"]
[vrodriiguez@contenedores dockerfile-test]$ docker image inspect dockerfile-demo --format '{{json .Config.Volumes}}'
{"data":{}}
[vrodriiguez@contenedores dockerfile-test]$
```